MEDICAL IMAGE CLASSIFICATION USING CNN

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1. Introduction

This project is based on Medical Image Classification using Deep Learning. The system uses a Convolutional Neural Network (CNN) to classify chest X-ray images into Normal and Pneumonia categories.

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2. Objective

The main objective of this project is to build a model that can:

- Detect Pneumonia from chest X-ray images
- Classify medical images automatically
- Assist in early disease detection

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3. Dataset Description

The dataset used is the Chest X-ray dataset.

It contains two classes:

- NORMAL
- PNEUMONIA

The dataset is divided into training and testing folders.

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4. Tools Used

- Python
- TensorFlow / Keras
- NumPy
- Pandas

- Matplotlib
- Jupyter Notebook

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5. Methodology

Step-by-step process:

1. Load dataset using ImageDataGenerator
2. Resize images to 150x150
3. Normalize pixel values
4. Build CNN model
5. Train model on dataset
6. Test model on unseen data
7. Evaluate accuracy
8. Save trained model

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6. CNN Model Architecture

The CNN model includes:

- Convolution Layer (32 filters)
- Max Pooling Layer
- Convolution Layer (64 filters)
- Max Pooling Layer
- Flatten Layer
- Dense Layer (128 neurons)
- Output Layer (Sigmoid activation)

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7. Results

- Training Accuracy: 98.5%
- Testing Accuracy: 75%

The model successfully classifies chest X-ray images into Normal and Pneumonia.

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8. Conclusion

The project shows how deep learning can be used in medical image classification. CNN helps in detecting patterns in X-ray images and can support medical diagnosis.

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9. Future Scope

- Improve accuracy using advanced CNN models
- Use larger datasets
- Add data augmentation
- Deploy as web application

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10. Author

Name: Maseera Kowsar